Purity of Pure and Fundamental Propositions

An explanatory proof of Jacob Lettie's theorem

11 November 2026

Can purity itself be pure if there is a fundamental proposition? Under Bacon's full Quantified Logical Necessity, Lettie's theorem answers Goodman's question negatively. The argument needs only a small part of the theory, and it does not assume that fundamentality is pure.

The theorem and its proof are due to Jacob Lettie, working with the Purity of Pure team, a group of Claude Opus 5.5 sessions.[1] This note explains the construction and proves the exact finite-premise result formalized in the Isabelle files supplied by Lettie. The correspondence is recorded in the appendix. Those files are unchanged.

1 From fundamentality to a universal proposition

Propositions have type t, and $u = t \to t$ is the type of unary propositional operators. Write $\pi = \mathrm{Pure}_u$. Thus the instance of Purity of Pure at issue is

$$\mathrm{PP}_u := \mathrm{Pure}_{u\to t}(\pi).$$

It says that the predicate applying to pure unary operators is itself pure. This is different from the bottom-type instance $\mathrm{Pure}_{t\to t}(\mathrm{Pure}_t)$.

Throughout, x, y, z, s, r range over propositions and X, Y over operators of type u. We use the definitions

$$\Box p := (p = \top), \qquad \Diamond p := {\sim}\Box{\sim}p.$$

Following Lettie, put

$$R(x,y) := \forall Y^u \big(\pi(Y) \to (Y(y) \to \Diamond Y(x))\big), \tag{1}$$

$$\mathrm{Uni}(x) := \forall y^t\, R(x,y). \tag{2}$$

The relation $R(x,y)$ says that x can have every pure property that y has. Here *property* means a propositional operator of type u. We call x *universal* if this holds for every proposition y. Different properties may be possible at different possibilities; universality does not require their joint possible instantiation.

[1] The source notes are *If a Proposition Is Fundamental, Purity Is Not Pure* and *Weaker Theories That Still Refute Purity of Pure*, dated 9 November 2026.

This notion extracts what the proof needs from fundamentality. Unary Recombination says

$$(\mathrm{Pure}(X)\,\&\,\mathrm{Fun}(r)) \to \big(\Box X(r) \to \forall q\, X(q)\big).$$

Logical Purity says that closed logical terms are pure. Application says that a pure operator applied to a pure argument has a pure value.

Proposition (Fundamental propositions are universal)**.** *Given Logical Purity, Application, and unary Recombination,* $\forall r(\mathrm{Fun}(r) \to \mathrm{Uni}(r))$.

Proof. Let r be fundamental. Fix a proposition y and a pure operator Y such that $Y(y)$. The operator $X := \lambda z.{\sim}Y(z)$ is pure: the closed logical term $\lambda W^u.\lambda z.{\sim}W(z)$ is pure, and Application applies it to the pure Y.

If $\Box{\sim}Y(r)$, Recombination at X gives $\forall q\,{\sim}Y(q)$, contradicting $Y(y)$. Hence ${\sim}\Box{\sim}Y(r)$, which is $\Diamond Y(r)$. Since y and Y were arbitrary, $\mathrm{Uni}(r)$. ∎

It therefore suffices to show that PP, together with suitable purity principles, rules out every universal proposition. From this point onward the proof contains no Fun.

2 The construction and the idea of the proof

We work over H + PE: classical higher-order logic H with Leibniz identity, $\beta\eta$-conversion, and Propositional Equivalence at type t. The latter rule takes a *theorem* $A \leftrightarrow B$ to $A = B$. Modalized Functionality is not assumed.

The resulting base logic proves the S4 laws

$$\mathrm{K}: \quad \Box(A \to B) \to (\Box A \to \Box B),$$

$$\mathrm{T}: \quad \Box A \to A, \qquad\qquad 4: \quad \Box A \to \Box\Box A,$$

Boolean identities, and

$$\Box(p \leftrightarrow q) \to p = q.$$

It is also closed under Necessitation. These rules apply to base theorems, including theorems containing the constants Pure_σ. They do *not* authorize necessitating consequences of additional purity premises. We will mark the point where those premises enter.

Define

(3) $C(z) := {\sim}R(z, {\sim}z) \,\&\, {\sim}R({\sim}z, z),$

(4) $F(z) := \Box C(z),$

(5) $Q(z) := F(z) \to z,$

(6) $P(z) := {\sim}z \,\&\, {\sim}F(z), \qquad P'(z) := z \,\&\, {\sim}F(z).$

Each expression denotes a term built from logical vocabulary and π. In particular, F, Q, P, P' have type u.

Lettie calls z *firm* when $F(z)$. Necessarily, neither z nor its negation can have every pure property of the other. Firmness is thus a modal separation condition, and its definition is symmetric in z and ${\sim}z$.

The operator Q is designed to fix firm propositions. This requires an identity, $Q(z) = z$, not merely a biconditional; Lemma B will establish it. The two tests P and P' detect, respectively, false and true propositions that are not firm. Their role is to force *every* value of Q to be firm.

For an operator Y, write $Y \circ Q := \lambda z.\, Y(Q(z))$. Here is the plan. The purity premises and PP will give

(7) $\forall x\, F(Q(x)), \qquad \forall Y\big(\pi(Y) \to \pi(Y \circ Q)\big).$

Suppose that s is universal, and set $e = Q(s)$. The first fact makes e firm. Symmetry then makes ${\sim}e$ firm, so $Q({\sim}e) = {\sim}e$. The second fact lets us apply universality of s to the pure operator $Y \circ Q$. It follows that every pure property of ${\sim}e$ is possible for e, giving $R(e, {\sim}e)$. But firmness of e rules out exactly that relation.

The remaining work is to establish the two facts in (7) without applying Necessitation above the added premises.

3 The nine purity premises

We state the finite assumptions explicitly. Abstracting the sole nonlogical constant π from the terms above gives closed logical terms M, M', M'' such that

(8) $M(\pi) = P, \qquad M'(\pi) = P', \qquad M''(\pi) = \lambda Y^{u}.\, Y \circ Q.$

Their types are $(u \to t) \to u$, $(u \to t) \to u$, and $(u \to t) \to (u \to u)$, respectively. More explicitly, let F_c, Q_c be the terms obtained by replacing π uniformly by a variable $c^{u \to t}$. Then

$$M := \lambda c.\lambda z.(\sim z \,\&\, \sim F_c(z)),$$
$$M' := \lambda c.\lambda z.(z \,\&\, \sim F_c(z)),$$
$$M'' := \lambda c.\lambda Y.\lambda z.Y(Q_c(z)).$$

Equation (8) follows by β-conversion.

Write

$$\mathrm{App}(\sigma, \tau) := \forall X^{\sigma \to \tau} \forall y^{\sigma} \big((\mathrm{Pure}(X) \,\&\, \mathrm{Pure}(y)) \to \mathrm{Pure}(X(y))\big).$$

We use the following nine ordinary premises:

$$\mathrm{W1} : \mathrm{Pure}(M), \qquad \mathrm{W2} : \mathrm{Pure}(M'), \qquad \mathrm{W3} : \mathrm{Pure}(M'').$$

$$\mathrm{W4} : \mathrm{App}(u \to t, u), \quad \mathrm{W5} : \mathrm{App}(u \to t, u \to u),$$
$$\mathrm{W6} : \mathrm{App}(u, t), \qquad \mathrm{W7} : \mathrm{App}(u, u),$$

and

$$\mathrm{W8} : \mathrm{Pure}_t(\pi(P)) \to (\pi(P) \to \Box\pi(P)),$$
$$\mathrm{W9} : \mathrm{Pure}_t(\pi(P')) \to (\pi(P') \to \Box\pi(P')).$$

The first three are instances of Logical Purity, the next four are typed Application instances, and the last two are instances of zeroary Exhaustion,

$$\mathrm{Pure}_t(a) \to (a \to \Box a).$$

None of these premises is assumed necessary.

Theorem 1 (Lettie)**.** *Over* H + PE*, the premises W1–W9 and* PP_u *imply* $\sim\exists x\, \mathrm{Uni}(x)$*.*

Equivalently, let W *be* H+PE *with W1–W9 and the tenth ordinary premise* $\mathrm{W10} : \exists x\, \mathrm{Uni}(x)$*. Then*

$$\mathrm{W} \vdash \sim\mathrm{PP}_u.$$

We prove the theorem in three stages: establish the logical properties of Q, use the purity premises to obtain (7), and derive the contradiction from universality.

4 The logical part of the construction

All results in this section are theorems of H + PE. They mention π but make no assumption about what is pure.

Lemma A (Firmness persists)**.** $F(x) \to \Box F(x)$.

Proof. By definition $F(x) = \Box C(x)$, so axiom 4 gives the result. ∎

Lemma B (Q fixes firm propositions)**.** $F(x) \to Q(x) = x$.

Proof. The formula $F(x) \to (Q(x) \leftrightarrow x)$ is a tautology. Necessitating this base theorem and using K gives $\Box F(x) \to \Box(Q(x) \leftrightarrow x)$. Assume that $F(x)$. Lemma A supplies $\Box F(x)$, so $\Box(Q(x) \leftrightarrow x)$. The base-logical identity principle $\Box(p \leftrightarrow q) \to p = q$ now gives $Q(x) = x$. ∎

Lemma C (Symmetry)**.** $F(\sim x) = F(x)$, *and* $P'(z) = P(\sim z)$.

Proof. Expand $C(\sim x)$ and use the Boolean identity $\sim\sim x = x$. By Leibniz's law,

$$C(\sim x) = \sim R(\sim x, x) \,\&\, \sim R(x, \sim x) = C(x).$$

The second identity is commutativity of conjunction, a base theorem obtained by PE. Applying $\lambda p.\Box p$ to the identity gives $F(\sim x) = F(x)$.

For the second claim, Leibniz's law and the same double-negation identity give

$$P(\sim z) = \sim\sim z \,\&\, \sim F(\sim z) = z \,\&\, \sim F(z) = P'(z).$$

∎

The choice $Q(x) = F(x) \to x$ has a useful consequence: a false value of Q cannot be nonfirm. This is why the first test P matters.

Lemma D. $\sim Q(x) \to F(Q(x))$.

Proof. Assume that $\sim Q(x)$, that is, $\sim(F(x) \to x)$. Then $F(x)$, and Lemma B gives $Q(x) = x$. Substitution into $F(x)$ yields $F(Q(x))$. ∎

Lemma E (The first test necessarily fails on Q's values)**.** $\Box \sim P(Q(x))$.

Proof. Suppose that $P(Q(x))$. Then $\sim Q(x)$ and $\sim F(Q(x))$. Lemma D turns the first conjunct into $F(Q(x))$, contradicting the second. Thus $\sim P(Q(x))$ is a base theorem. Necessitation gives the displayed result. ∎

The next lemma uses both tests. Its hypotheses are necessary purity statements; the lemma itself is still a theorem of the base logic.

Lemma F (Necessary purity of the tests suffices)**.**

$$\big(\Box\pi(P)\,\&\,\Box\pi(P')\big) \to F(Q(x)).$$

Proof. Put $q := Q(x)$. We first establish the unboxed implication

$$\big(\pi(P)\,\&\,\pi(P')\big) \to C(q). \tag{9}$$

Assume that $\pi(P)$ and $\pi(P')$, and suppose, for a contradiction, that $C(q)$ is false. Thus

$$R(q,{\sim}q) \vee R({\sim}q,q).$$

The proposition q is not firm: $F(q) = \Box C(q)$ would imply $C(q)$ by T. Since ${\sim}F(q)$, the contrapositive of Lemma D gives q. Consequently,

$$P'(q), \qquad P({\sim}q).$$

The first follows from $q\,\&\,{\sim}F(q)$. For the second, use ${\sim}{\sim}q = q$ and $F({\sim}q) = F(q)$ from Lemma C.

If $R(q,{\sim}q)$, instantiate its defining quantifier at the pure operator P. Since $P({\sim}q)$, it gives $\Diamond P(q)$. If instead $R({\sim}q,q)$, instantiate at the pure operator P'. Since $P'(q)$, it gives $\Diamond P'({\sim}q)$. But

$$P'({\sim}q) = P({\sim}{\sim}q) = P(q)$$

by Lemma C and double negation. Again $\Diamond P(q)$.

In both cases this contradicts $\Box{\sim}P(q)$, supplied by Lemma E. Discharging the temporary assumptions proves (9) in H + PE. Necessitation and K therefore give

$$\big(\Box\pi(P)\,\&\,\Box\pi(P')\big) \to \Box C(q).$$

Since $\Box C(q) = F(Q(x))$, this is the required implication. ∎

5 The purity argument

To prove Theorem 1, we now use W1–W9. The key step makes a true assertion of purity itself pure, so that Exhaustion applies.

Lemma G. *The premises W1–W9 and* PP_u *give*

$$\Box\pi(P), \qquad \Box\pi(P'), \qquad \forall x\, F(Q(x)), \qquad \forall Y\big(\pi(Y) \to \pi(Y \circ Q)\big).$$

Proof. First consider P. The term M is pure by W1, and π is pure by PP_u. Application at W4, together with $M(\pi) = P$, gives

$$\pi(P).$$

Now π and P are both pure. Application at W6 gives

$$\mathrm{Pure}_t(\pi(P)).$$

Thus $\pi(P)$ is both true and pure. Exhaustion at W8 yields

$$\Box\pi(P).$$

The same argument, using W2 and W9 in place of W1 and W8, yields $\Box\pi(P')$. Lemma F then gives $F(Q(x))$ for every x.

For composition, W3 makes M'' pure. With PP_u and Application at W5, this makes $M''(\pi) = \lambda Y.\, Y \circ Q$ pure. If Y is pure, Application at W7 therefore makes $Y \circ Q$ pure. Generalizing Y proves the last conclusion. ∎

Exhaustion has supplied necessary purity without necessitating the premises. PP supplies the purity of π; Application lets that purity be used a second time, to make $\pi(P)$ itself a pure proposition. This second use of the purity of π is what allows Exhaustion to apply.

6 The contradiction

Proof of Theorem 1. Assume that PP_u and W1–W9 hold, and suppose that s is universal. Put $e := Q(s)$. Lemma G makes e firm. By symmetry $\sim e$ is firm too, so Lemma B gives

$$Q(\sim e) = \sim e. \tag{10}$$

We show that $R(e, \sim e)$. Let Y be a pure operator such that $Y(\sim e)$. By Lemma G, $Y \circ Q$ is pure, and (10) gives

$$(Y \circ Q)(\sim e) = Y(Q(\sim e)) = Y(\sim e).$$

Thus $Y \circ Q$ holds of $\sim e$. Since s is universal, $R(s, \sim e)$. Apply its defining quantifier to $Y \circ Q$.

We obtain $\Diamond(Y \circ Q)(s)$, which by β-conversion is

$$\Diamond Y(Q(s)), \quad \text{that is,} \quad \Diamond Y(e).$$

Since Y was arbitrary, this proves $R(e, {\sim}e)$.

But $F(e) = \Box C(e)$ gives $C(e)$ by T, and its first conjunct is ${\sim}R(e, {\sim}e)$. This is a contradiction. Therefore no universal proposition exists under PP_u and W1–W9.

Adding W10 gives a contradiction. Since the premises are sentences and Necessitation has been applied only to base theorems, the ordinary deduction theorem discharges PP_u. Hence $\mathrm{W} \vdash {\sim}\mathrm{PP}_u$. ■

Combining this theorem with the proposition in Section 1 gives the negative answer to Goodman's full-QLN question. Among the QLN principles, only unary Recombination and two instances of zeroary Exhaustion are needed. Purity of Fun, uniqueness of the fundamental proposition, general Persistence, Modalized Functionality, and higher-arity QLN are not used.

If zeroary instances are not included explicitly in QLN, unary Exhaustion and a fundamental proposition suffice here. Given a true pure proposition a, the constant operator $\lambda q.a$ is pure by Logical Purity and Application. It holds of every proposition. Unary Exhaustion at a fundamental r therefore gives $\Box a$.

Without Exhaustion, the two particular Persistence premises $\pi(P) \to \Box\pi(P)$ and $\pi(P') \to \Box\pi(P')$ would also supply the needed boxes. Without either source of necessary purity, the ordinary-premise Recombination-only question remains open.

Appendix: exact statement and verification correspondence

The main theorem is `w_not_pp` in Lettie's Isabelle development. It is a closed theorem of `U_proves`, a calculus for H with PE, modus ponens, Generalization, and Instantiation. There is no rule for added premises. W1–W10 occur as the ten antecedents of

$$\mathrm{H} + \mathrm{PE} \vdash \mathrm{W1} \to (\mathrm{W2} \to \cdots \to (\mathrm{W10} \to {\sim}\mathrm{PP}_u) \cdots).$$

The displayed implications associate to the right. Consequently the formalization cannot apply PE or Necessitation to an antecedent as if it were an axiom. The theorem `w_not_pp_C` carries this same implication into the application's CEV calculus.

This note	Isabelle theorem or definition
Fundamentality gives universality	`u_uni_ctx`; existential consequence: `u_fun_ExUni`
R, Uni, F, Q, P, P'	`rcR`, `rcUni`, `rcF`, `rcQ`, `rcP`, `rcPp`
M, M', M''	`rc_P`, `rc_Pp`, `rc_W`
Lemma A	`nup_lemA`
Lemma B	`nup_lemB`
Lemma C	`nup_lemC`, `nup_ident`
Lemma D	`nup_lemD`
Lemma E	`nup_cor`
Lemma F	`nup_step2`
Lemma G	`nup_step3`
The final contradiction	`nup_step4`
No Universal Proposition	`nup_NUP_U`, `u_S`
The ten-premise refutation	`w_not_pp`, `w_not_pp_C`
The full ordinary-premise package	`cfp2`, `cfp2_thm`

The definitions and the W1–W10 inventory in this note follow the Isabelle statement, using the conjunctive $F(z) = \Box(\sim R(z, \sim z)\ \&\ \sim R(\sim z, z))$. Lettie's main note and Metamath certificate instead write $F(z) = \Box\sim(R(z, \sim z) \vee R(\sim z, z))$. Their values at each argument are identical by PE. This does not by itself identify the two operators under λ-abstraction over H + PE. The corresponding abstractors in W1–W3 must therefore be read with the chosen definition; this note does not silently interchange the two finite premise lists.

Lettie's files in `theories/lettie_pp/` are included unmodified and with permission. The repository records a successful Isabelle2026 build/export and an audit of 17 principal endpoints with no oracles, residual hypotheses, or unresolved unification pairs. The complete mapping is in `docs/LETTIE_PP_CORRESPONDENCE.md`.

The repository also records verification of Lettie's separately supplied Metamath certificate `w-not-pp` on 11 November 2026. Three verifiers were used: `metamath.exe`, `checkmm`, and `mmverify.py`. Its dependency trace uses only typing, substitution and conversion, the propositional and quantifier rules, and PE, not Modalized Functionality or the file's purity or QLN axioms. That inspection covered the statements of the rules used, not the entire axiom base. The Metamath file is not distributed in this repository. This explanatory revision changes no Isabelle statement or proof.